# National Neighborhood Data Archive (NaNDA): Personal Care Services and Laundromats by Census Tract, United States, 2003-2017

openICPSR-115981
nanda_pclaund_tract_2003-2017_01P.dta
nanda_pclaund_tract_2003-2017_01P.csv
nanda_pclaund0317T_01P.sas7bdat

# Overview and Data Dictionary

Documentation Version: 1.1
Last updated: 12/7/2020

## Dataset Overview

### Description

This dataset contains measures of the number and per capita density of personal care services (such as barber shops, hair and nail salons, and spas) and laundromats per United States census tract from 2003 through 2017.

### Principal Investigators


- Michael Esposito, University of Michigan Institute for Social Research
- Mao Li, University of Michigan Institute for Social Research
- Jessica Finlay, University of Michigan Institute for Social Research
- Iris Gomez-Lopez, University of Michigan Institute for Social Research
- Anam Khan, University of Michigan Institute for Social Research
- Philippa Clarke, University of Michigan Institute for Social Research
- Megan Chenoweth, University of Michigan Institute for Social Research

# Funding Sources


- United States Department of Health and Human Services. Administration for Community Living. National Institute on Disability, Independent Living, and Rehabilitation Research (90RTHF0001)
- United States Department of Health and Human Services. National Institutes of Health. National Institute on Aging (RF1-AG-057540)


# Data Sources

The North American Industry Classification System (NAICS) was used to identify organizations of interest. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

# Coverage

The dataset contains one observation per census tract per year in the United States, including Puerto Rico and the US Virgin Islands. Other US island territories are excluded.

# Methodology

This dataset highlights the concentration of certain types of personal care services—including barber shops, salons, spas, and weight loss centers—and laundromats per census tract in the United States. These businesses, along with more common gathering places (e.g. coffee shops, churches, senior centers), function as common “third places” in the United States (Finlay et al., 2019). Third places are locations other than home (first place) and work/school (second place) where people gather, spend time, and casually interact with one another; they are social infrastructure vital to civic life (Klinenberg, 2018). Prior research has shown that their presence in a neighborhood fosters foster social interaction, community building, and individual well-being (Finlay et al., 2018). They can also serve as locations for health promotion and literacy initiatives among minority populations and in high-poverty areas (Baron & Trujillo, 2019, Linnan et al., 2014, Murphy et al., 2017).

Our research team identified the following NAICS codes of interest:

- 812111: barber shops (men's hair stylists).
- 812112: beauty salons (unisex and women's hair stylists).
- 812113: nail salons.
- 812191: diet and weight loss centers.
- 812199: other personal care services, which include spas, saunas, massage parlors, tattoo parlors, and tanning stations.
- 812310: coin-operated laundries and dry cleaners (laundromats).

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment's census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010. Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017. Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other

NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019) and Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all personal care services and laundromats per tract in each year.

## Usage Note

Data users interested in third places and community resources might find further useful data in these additional NaNDA datasets might find further useful data in these additional NaNDA datasets:

- Arts, entertainment, and recreation organizations by census tract, United States, 2003-2017 (Finlay et al., 2020a), which includes gyms, libraries, and arcades, among others.
- Eating and drinking places by census tract, United States, 2003-2017 (Esposito et al., 2020), which includes coffee shops, fast food restaurants, and bars.
- Grocery stores by census tract, United States, 2003-2017 (Finlay et al., 2020b), which includes warehouse clubs.
- Post offices and banks by census tract, United States, 2003-2017 (Finlay et al., 2020c).
- Religious, civic, and social organizations by census tract, United States, 2003-2017 (Finlay et al., 2020d).
- Retail establishments by census tract, United States, 2003-2017 (Finlay et al., 2020e).
- Social services by census tract, United States, 2003-2017 (Finlay et al., 2020f), which includes senior centers.

# Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: [https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017](https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017).

| Variable | Type | Obs | Unique | Mean | Min | Max | Label |
|---|---|---|---|---|---|---|---|
| tract_fips10 | string | 1036462 | 74033 | . | . | . | Census tract FIPS code (2010 TIGER/Line shapefiles) |
| year | int | 1036462 | 14 | 2009.571 | 2003 | 2017 | Year |
| population | float | 1020881 | 250271 | 4208.894 | 0 | 65528 | Tract population |
| aland10 | float | 1022434 | 72534 | 48.07265 | 0 | 85425.73 | Tract land area, square miles |
| count_812111 | int | 1036462 | 25 | 0.675992 | 0 | 25 | # barber shops |
| count_sales_812111 | int | 1036462 | 25 | 0.6315987 | 0 | 25 | # barber shops w/ sales>0 |
| count_emp_812111 | int | 1036462 | 19 | 0.391915 | 0 | 19 | # barber shops w/ 2+ employees |
| popden_812111 | float | 1015151 | 152760 | 31.09827 | -1 | 1.17E+07 | # barber shops per 1000 people |
| popden_sales_812111 | float | 1015142 | 146991 | 31.06969 | -1 | 1.17E+07 | # barber shops per 1000 people w/ sales>0 |
| popden_emp_812111 | float | 1015108 | 109359 | 21.50983 | -1 | 1.17E+07 | # barber shops per 1000 people w/ 2+ employees |
| aden_812111 | float | 1017996 | 73467 | 1.125472 | 0 | 268.7466 | # barber shops per sq mile |
| aden_sales_812111 | float | 1017996 | 71864 | 1.04075 | 0 | 268.7466 | # barber shops per sq mile w/ sales>0 |
| aden_emp_812111 | float | 1017996 | 57389 | 0.6885337 | 0 | 268.7466 | # barber shops per sq mile w/ 2+ employees |
| count_812112 | int | 1036462 | 141 | 4.716594 | 0 | 181 | # beauty salons |
| count_sales_812112 | int | 1036462 | 121 | 4.352518 | 0 | 170 | # beauty salons w/ sales>0 |
| count_emp_812112 | int | 1036462 | 95 | 3.041134 | 0 | 132 | # beauty salons w/ 2+ employees |
| popden_812112 | float | 1015511 | 361868 | 33.32525 | -1 | 1.17E+07 | # beauty salons per 1000 people |
| popden_sales_812112 | float | 1015487 | 353063 | 32.99986 | -1 | 1.17E+07 | # beauty salons per 1000 people w/ sales>0 |
| popden_emp_812112 | float | 1015419 | 309372 | 32.75496 | -1 | 1.17E+07 | # beauty salons per 1000 people w/ 2+ employees |
| aden_812112 | float | 1017996 | 292064 | 6.693071 | 0 | 1312.099 | # beauty salons per sq mile |
| aden_sales_812112 | float | 1017996 | 285379 | 6.145752 | 0 | 1217.411 | # beauty salons per sq mile w/ sales>0 |
| aden_emp_812112 | float | 1017996 | 250720 | 4.479826 | 0 | 946.8754 | # beauty salons per sq mile w/ 2+ employees |
| count_812113 | int | 1036462 | 21 | 0.7409206 | 0 | 20 | # nail salons |

| Variable | Type | Obs | Unique | Mean | Min | Max | Label |
|---|---|---|---|---|---|---|---|
| count_sales_812113 | int | 1036462 | 21 | 0.6790331 | 0 | 20 | # nail salons w/ sales>0 |
| count_emp_812113 | int | 1036462 | 17 | 0.5308569 | 0 | 16 | # nail salons w/ 2+ employees |
| popden_812113 | float | 1015175 | 149298 | 0.2132341 | -1 | 5864.635 | # nail salons per 1000 people |
| popden_sales_812113 | float | 1015169 | 141275 | 0.195997 | -1 | 5864.635 | # nail salons per 1000 people w/ sales>0 |
| popden_emp_812113 | float | 1015141 | 124336 | 0.1549634 | -1 | 5864.635 | # nail salons per 1000 people w/ 2+ employees |
| aden_812113 | float | 1017996 | 78568 | 1.105349 | 0 | 230.8943 | # nail salons per sq mile |
| aden_sales_812113 | float | 1017996 | 77225 | 1.01283 | 0 | 230.8943 | # nail salons per sq mile w/ sales>0 |
| aden_emp_812113 | float | 1017996 | 69020 | 0.7979881 | 0 | 230.8943 | # nail salons per sq mile w/ 2+ employees |
| count_812191 | int | 1036462 | 7 | 0.0489405 | 0 | 6 | # weight loss centers |
| count_sales_812191 | int | 1036462 | 6 | 0.044294 | 0 | 5 | # weight loss centers w/ sales>0 |
| count_emp_812191 | int | 1036462 | 6 | 0.0322376 | 0 | 6 | # weight loss centers w/ 2+ employees |
| popden_812191 | float | 1015064 | 24698 | 0.019185 | -1 | 763.4285 | # weight loss centers per 1000 people |
| popden_sales_812191 | float | 1015063 | 22976 | 0.0169199 | -1 | 500 | # weight loss centers per 1000 people w/ sales>0 |
| popden_emp_812191 | float | 1015058 | 17481 | 0.0146557 | -1 | 763.4285 | # weight loss centers per 1000 people w/ 2+ employees |
| aden_812191 | float | 1017996 | 7884 | 0.0542993 | 0 | 54.85985 | # weight loss centers per sq mile |
| aden_sales_812191 | float | 1017996 | 7812 | 0.0493702 | 0 | 54.85985 | # weight loss centers per sq mile w/ sales>0 |
| aden_emp_812191 | float | 1017996 | 6815 | 0.0350986 | 0 | 49.82759 | # weight loss centers per sq mile w/ 2+ employees |
| count_812199 | int | 1036462 | 35 | 0.9490594 | 0 | 46 | # other personal care services |
| count_sales_812199 | int | 1036462 | 35 | 0.8567386 | 0 | 44 | # other personal care services w/ sales>0 |
| count_emp_812199 | int | 1036462 | 23 | 0.5230129 | 0 | 27 | # other personal care services w/ 2+ employees |
| popden_812199 | float | 1015227 | 177922 | 0.602519 | -1 | 119069 | # other personal care services per 1000 people |
| popden_sales_812199 | float | 1015217 | 167591 | 0.5373433 | -1 | 119069 | # other personal care services per 1000 people w/ sales>0 |
| popden_emp_812199 | float | 1015169 | 122459 | 0.2090846 | -1 | 29767.25 | # other personal care services per 1000 people w/ 2+ employees |
| aden_812199 | float | 1017996 | 105714 | 1.059374 | 0 | 325.6372 | # other personal care services per sq mile |
| aden_sales_812199 | float | 1017996 | 102878 | 0.9566637 | 0 | 325.6372 | # other personal care services per sq mile w/ sales>0 |
| aden_emp_812199 | float | 1017996 | 82824 | 0.5793056 | 0 | 232.5929 | # other personal care services per sq mile w/ 2+ employees |
| count_812310 | int | 1036462 | 13 | 0.320855 | 0 | 18 | # laundromats |
| count_sales_812310 | int | 1036462 | 13 | 0.2995276 | 0 | 18 | # laundromats w/ sales>0 |
| count_emp_812310 | int | 1036462 | 13 | 0.2804251 | 0 | 18 | # laundromats w/ 2+ employees |
| popden_812310 | float | 1015108 | 97940 | 0.0955099 | -1 | 1526.857 | # laundromats per 1000 people |
| popden_sales_812310 | float | 1015103 | 93903 | 0.089297 | -1 | 1526.857 | # laundromats per 1000 people w/ sales>0 |

| Variable | Type | Obs | Unique | Mean | Min | Max | Label |
|---|---|---|---|---|---|---|---|
| popden_emp_812310 | float | 1015106 | 88690 | 0.0849544 | -1 | 1526.857 | # laundromats per 1000 people w/ 2+ employees |
| aden_812310 | float | 1017996 | 40055 | 0.8021589 | 0 | 203.0938 | # laundromats per sq mile |
| aden_sales_812310 | float | 1017996 | 39352 | 0.7436774 | 0 | 203.0938 | # laundromats per sq mile w/ sales>0 |
| aden_emp_812310 | float | 1017996 | 37181 | 0.7151393 | 0 | 203.0938 | # laundromats per sq mile w/ 2+ employees |

# References

Baron, S., & Trujillo, J. (2019). Expanding our reach, using our strengths: From laundromats to manufactured housing communities. *Childhood Education 29*(6), 56-58. https://doi.org/10.1080/00094056.2019.1689067

Clarke, P. J., & Melendez, R. (2019). National Neighborhood Data Archive (NaNDA): Neighborhood socioeconomic and demographic characteristics of census tracts, United States, 2000-2010 (V1) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E111107V1

Esposito, M., Li, M., Finlay, J. M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., and Chenoweth, M. M. (2020). National Neighborhood Data Archive (NaNDA): Eating and drinking places by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E115404V2

Finlay, J. M., Gaugler, J. E., & Kane, R. L. (2018). Ageing in the margins: Expectations of and struggles for “a good place to grow old” among low-income older Minnesotans. *Ageing and Society*, 1-25. https://doi.org/10.1017/S0144686X1800123X

Finlay, J. M., Esposito, M., Kim, M. H., Gomez-Lopez, I. N., & Clarke, P. J. (2019). Closure of “third places”? Exploring potential consequences for collective health and wellbeing. *Health & Place, 60*, 102225. https://doi.org/10.1016/j.healthplace.2019.102225

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020a). National Neighborhood Data Archive (NaNDA): Arts, entertainment, and recreation organizations by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E115543V2

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020b). National Neighborhood Data Archive (NaNDA): Grocery stores by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E123001V1

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020c). National Neighborhood Data Archive (NaNDA): Post offices and banks by census tract, United States, 2003-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E128281V1

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020d). National Neighborhood Data Archive (NaNDA): Religious, civic, and social organizations by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E115967V1

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020e). National Neighborhood Data Archive (NaNDA): Retail establishments by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E115972V2

Finlay, J. M., Li, M., Esposito, M., Gomez-Lopez, I. N., Khan, A. M., Clarke, P. J., & Chenoweth, M. M. (2020f). National Neighborhood Data Archive (NaNDA): Social services by census tract, United States, 2003-2017 (V2) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E117163V2

Gomez-Lopez, I. N., Clarke, P. J., Hill, A. B., Romero, D. M., Goodspeed, R., Berrocal, V. J., Vinod Vydiswaran, V.G, & Veinot, T. C. (2017). Using social media to identify sources of healthy food in urban neighborhoods. *Journal of Urban Health, 94*(3), 429–36. https://doi.org/10.1007/s11524-017-0154-1

Klinenberg, E. (2018). *Palaces for the people: How social infrastructure can help fight inequality, polarization, and the decline of civic life.* Broadway Books.

Linnan, L. A., D'Angelo, H., & Harrington, C. A. (2014). A literature synthesis of health promotion research in salons and barbershops. *American Journal of Preventive Medicine 47*(1), 77-85. https://doi.org/10.1016/j.amepre.2014.02.007

Melendez, R., Clarke, P. J., Khan, A. M., Gomez-Lopez, I. N., Li, M., & Chenoweth, M. (2020). National Neighborhood Data Archive (NaNDA): Socioeconomic status and demographic characteristics of census tracts, United States, 2008-2017 (V1) [Data set]. Inter-university Consortium for Political and Social Research. https://doi.org/10.3886/E120462V1

Murphy, A. B., Moore, N. J., Wright, M., Gipson, J., Keeter, M., Cornelious, T., Reed, D., Russell, J., Watson, K. S., & Murray, M. (2017). Alternative locales for the health promotion of African American men: A survey of African American men in Chicago barbershops. *Journal of Community Health 42*, 139-146. https://doi.org/10.1007/s10900-016-0240-4

United States Census Bureau (2010). TIGER/Line shapefiles, 2010 census tracts (2010 version) [Data set]. https://www2.census.gov/geo/tiger/TIGER2010/TRACT/2010/tl_2010_us_tract10.zip.

United States Census Bureau (2017). North American Industry Classification System (2017 version) [Data set]. https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017

Walls & Associates (2017). National Establishment Time Series (NETS) database.